Andrew Hallam

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RESEARCH INTERESTS

Quantum many-body physics, tensor networks and quantum information. Research includes out-of-equilibrium dynamics, quantum thermalisation and many-body scars; nonstabilizerness and quantum criticality; and spin-chain models of black-hole physics.

APPOINTMENTS

Quantum Fellow, EPCC, University of Edinburgh

Current appointment

Postdoctoral Fellow, University of Leeds

From 2020; previous appointment

Postdoctoral Fellow, Ghent University

2018-2020

EDUCATION

PhD in Physics, University College London

2015-2018 · Supervisor: Andrew Green

MRes in Delivering Quantum Technologies, University College London

2014-2015 · Distinction

MSci in Physics, University College London

2010-2014 · First class honours

TEACHING AND SUPERVISION

- Taught parts of **PHYS3003: Advanced Techniques in Theoretical Physics** and **PHYS5017M: Quantum Many-Body Physics** at the University of Leeds.
- Supervised several year-long Master's projects at Ghent University and the University of Leeds; co-supervised several PhD students at Leeds.

AWARD AND TECHNICAL EXPERIENCE

- **Mentorship Award for the Engineering and Physical Sciences**, University of Leeds, 2025.
- Significant programming experience in **Python, Julia and C++**; experience with **MATLAB and TensorFlow**.

Research experience

Projects undertaken during postdoctoral and postgraduate research.

UNIVERSITY OF LEEDS · POSTDOCTORAL RESEARCH

Supervisors: Jiannis Pachos and Zlatko Papić

- Used the entanglement spectrum of quantum ground states and the interaction distance to study one-dimensional interacting fermionic models.
- Built effective models of electrons on curved spacetime using one-dimensional spin chains.
- Studied non-equilibrium dynamics of systems that do not relax to thermal equilibrium, including quantum many-body scars and many-body localisation.

GHENT UNIVERSITY · POSTDOCTORAL RESEARCH

Supervisor: Jutho Haegeman

- Constructed a path integral summing over matrix product states in the continuum and at finite temperatures.
- Investigated the structure of the multiscale entanglement renormalization ansatz using matrix-product operator techniques.

UNIVERSITY COLLEGE LONDON · PHD

Tensor network descriptions of quantum entanglement in path integrals, thermalisation and machine learning.

Supervisor: Andrew Green

- Developed a method for simulating and compressing neural networks using tensor networks based on the multiscale entanglement renormalization ansatz.
- Explored fluctuations around the saddle point of a path integral constructed over matrix product states.
- Used a semiclassical description of quantum time evolution to study the relationship between classical chaos and quantum thermalisation.
- Combined tensor networks and the truncated spectrum approach to study a two-dimensional system created by suddenly coupling one-dimensional systems.

UNIVERSITY COLLEGE LONDON · MRES AND MSCI

MRes: Tensor network descriptions of out-of-equilibrium quantum systems. Supervisors: Andrew Green and Andrew James. Studied short-time behaviour after a two-dimensional quantum quench, using matrix product states in a cylindrical geometry.

MSci: Entanglement of Trapped Ions via Measurement. Supervisor: Dan Browne. Examined how common experimental imperfections affect the entanglement of atoms inside an optical cavity.

ACADEMIC SERVICE AND OUTREACH

- Organised the internal quantum theory seminar at Leeds and the Leeds-Loughborough-Nottingham non-equilibrium seminar: invited and hosted speakers, recorded talks, and maintained the website and YouTube channel.
- Helped organise UCL Physics and Astronomy's postgraduate conference **QCumber** in 2015 and 2016, and the **24th Symposium on Quantum Matter and High-Energy Physics** in 2022; arranged speakers, transport and accommodation.
- Helped deliver a hands-on quantum computing workshop for Year 12 and 13 pupils through Leeds' **Reach for Excellence** widening participation programme.

Publications and preprints

2025-2026 · Links lead to the published article and/or arXiv record.

01 Entanglement and apparent thermality in simulated black holes

Iason A. Sofos, **Andrew Hallam**, and Jiannis K. Pachos

SciPost Physics Core 9, 067 · 2026 · Publication · arXiv

02 Quantum many-body mixed phase space revealed by hybrid feedback control

Hang Dong, Jie Ren, **Andrew Hallam**, Han Wang, Zhengyi Cui, Yiren Zou, Junlin Wang, Hekang Li, Qiujiang Guo, Zhen Wang, Lei Ying, and Zlatko Papić

Nature Physics · 2026 · Publication · arXiv

03 Tunable quantum Mpemba effect in long-range interacting systems

Andrew Hallam, Matthew Yusuf, Aashish A. Clerk, Ivar Martin, and Zlatko Papić

Physical Review B 114, 125135 · 2026 · Publication · arXiv

04 Simulating Black Hole Thermality and Interior Scrambling on a Superconducting Quantum Processor

Ryan Smith, Ewan Forbes, Iason A. Sofos, **Andrew Hallam**, and Jiannis K. Pachos

arXiv:2608.19318 (preprint) · 2026 · arXiv

05 Spectral signatures of nonstabilizerness and criticality in infinite matrix product states

Andrew Hallam, Ryan Smith, and Zlatko Papić

Physical Review B 113, 245113 · 2026 · Publication · arXiv

06 Ergodicity breaking in matrix-product-state effective Hamiltonians

Andrew Hallam, Jared Jeyaretnam, and Zlatko Papić

arXiv:2603.26870 (preprint) · 2026 · arXiv

07 ScarFinder: A Detector of Optimal Scar Trajectories in Quantum Many-Body Dynamics

Jie Ren, **Andrew Hallam**, Lei Ying, and Zlatko Papić

PRX Quantum 6, 040332 · 2025 · Publication · arXiv

08 Nonstabilizerness in kinetically constrained Rydberg atom arrays

Ryan Smith, Zlatko Papić, and **Andrew Hallam**

Physical Review B 111, 245148 · 2025 · Publication · arXiv

09 Optimally scrambling chiral spin-chain with effective black hole geometry

Aiden Daniel, **Andrew Hallam**, Matthew D. Horner, and Jiannis K. Pachos

Scientific Reports 15, 9103 · 2025 · Publication · arXiv

10 Quantum teleportation between simulated binary black holes

Aiden Daniel, Tanmay Bhore, Jiannis K. Pachos, Chang Liu, and **Andrew Hallam**

arXiv:2503.10761 (preprint) · 2025 · arXiv

11 Stirring the false vacuum via interacting quantized bubbles on a 5,564-qubit quantum annealer

Jaka Vodeb, Jean-Yves Desaules, **Andrew Hallam**, Andrea Rava, Gregor Humar, Dennis Willsch, Fengping Jin, Madita Willsch, Kristel Michielsen, and Zlatko Papić

Nature Physics 21, 386-392 · 2025 · Publication · arXiv

Publications and preprints

Earlier work · Publications are listed in reverse chronological order.

12 Integrability breaking and bound states in Google's decorated XXZ circuits

Ana Hudomal, Ryan Smith, **Andrew Hallam**, and Zlatko Papić

PRX Quantum 5, 010316 · 2024 · Publication · arXiv

13 Embedding semiclassical periodic orbits into chaotic many-body Hamiltonians

Andrew Hallam, Jean-Yves Desaules, and Zlatko Papić

Physical Review Letters 131, 110401 · 2023 · Publication · arXiv

14 Bridging quantum criticality via many-body scarring

Aiden Daniel, **Andrew Hallam**, Jean-Yves Desaules, Ana Hudomal, Guo-Xian Su, Jad C. Halimeh, and Zlatko Papić

Physical Review B 107, 235108 · 2023 · Publication · arXiv

15 Exploring interacting chiral spin chains in terms of black hole physics

Ewan Forbes, Matthew D. Horner, **Andrew Hallam**, Joseph Barker, and Jiannis K. Pachos

Physical Review B 108, 245142 · 2023 · Publication · arXiv

16 Phase transitions in the classical simulability of open quantum systems

F. Azad, **Andrew Hallam**, J. Morley, and Andrew G. Green

Scientific Reports 13, 8866 · 2023 · Publication · arXiv

17 Chiral spin-chain interfaces exhibiting event-horizon physics

Matthew D. Horner, **Andrew Hallam**, and Jiannis K. Pachos

Physical Review Letters 130, 016701 · 2023 · Publication · arXiv

18 Tuning between continuous time crystals and many-body scars in long-range spin chains

Kieran Bull, **Andrew Hallam**, Zlatko Papić, and Ivar Martin

Physical Review Letters 129, 140602 · 2022 · Publication · arXiv

19 Emergence of gaussianity in the thermodynamic limit of interacting fermions

Gabriel Matos, **Andrew Hallam**, Aydin Deger, Zlatko Papić, and Jiannis K. Pachos

Physical Review B 104, L180408 · 2021 · Publication · arXiv

20 Entanglement compression in scale space: From the multiscale entanglement renormalization ansatz to matrix product operators

Karel Van Acoleyen, **Andrew Hallam**, Matthias Bal, Markus Hauru, Jutho Haegeman, and Frank Verstraete

Physical Review B 102, 165131 · 2020 · Publication · arXiv

21 The Lyapunov spectra of quantum thermalisation

Andrew Hallam, J. G. Morley, and Andrew G. Green

Nature Communications 10, 2708 · 2019 · Publication · arXiv

22 Hierarchical quantum classifiers

Edward Grant, Marcello Benedetti, Shuxiang Cao, **Andrew Hallam**, Joshua Lockhart, Vid Stojevic, Andrew G. Green, and Simone Severini

npj Quantum Information 4, 65 · 2018 · Publication · arXiv

23 Compact neural networks based on the multiscale entanglement renormalization ansatz

Andrew Hallam, Edward Grant, Vid Stojevic, Simone Severini, and Andrew G. Green

British Machine Vision Conference (BMVC 2018) · 2018 · Publication · arXiv

Presentations

Talks and posters at seminars, meetings and conferences.

TALKS

2026 **Understanding magic in critical many-body systems**
12th Northern Quantum Meeting, Google · 20 April

2026 **Understanding magic in critical many-body systems**
Tensor Network weekly seminar, Google · 6 April

2026 **Understanding magic in critical many-body systems**
University of Manchester · 1 April

2025 **Connecting many-body magic with quantum criticality with infinite matrix-product states**
Many-Body Quantum Magic (MBQM-2025), iQuS Seattle · 18-29 August

2024 **Non-stabilizerness in kinetically-constrained Rydberg atom arrays**
Bridging the physics and mathematics of quantum many-body chaos · 24-28 June

2023 **Quantum many-body physics**
White Rose Quantum Information and Technologies lectures · 13 December

2023 **Embedding semiclassical periodic orbits into chaotic many-body Hamiltonians**
IQTN workshop: Tensor networks for quantum many-body dynamics · 26-28 June

2022 **Tuning between continuous time crystals and many-body scars in long-range XYZ spin chains**
CMD29 · 21-26 August

2021 **Emergence of Gaussianity in the thermodynamic limit of interacting fermions**
Quantum, Atomic, and Molecular Physics (QuAMP 2021), online · 31 August-3 September

2019 **The Lyapunov spectrum of quantum thermalisation**
University of Freiburg · 2 July; Belgian Quantum Physics Initiative, Brussels · 2 May

2019 **Feynman path integrals over matrix product states**
Entanglement in Strongly Correlated Systems, Benasque · 24 February-9 March

2016 **Tensor networks and out-of-equilibrium dynamics in two dimensions**
QCumber, Cumberland Lodge · 16-18 June

POSTERS

2024 **Embedding semiclassical periodic orbits into chaotic many-body Hamiltonians**
IQTN 4th Plenary Meeting, Glasgow · 24-26 January

2021 **Interaction distance of the extended XYZ model**
Condensed Matter and Quantum Materials (CMQM 2021), online · 21-23 June

2018 **Compact Neural Networks based on the Multiscale Entanglement Renormalization Ansatz**
British Machine Vision Conference, Northumbria University · 3-6 September

2017 **The Lyapunov spectrum of thermalising quantum systems**
Frontiers in Condensed Matter Physics, Bristol · 11-12 January; Tensor17 School, Ghent · 6-10 November

2017 **Quantum quenches in two spatial dimensions**
Theory of Condensed Matter Group Annual Meeting, Warwick · 6 June; Frontiers in Condensed Matter Physics, Bristol · 9-10 January